

News and Perceptions 3

Paul Beaudry and Franck Portier

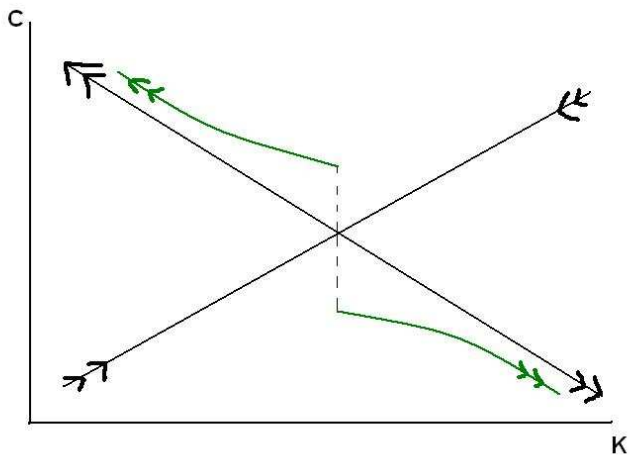
Econ 590

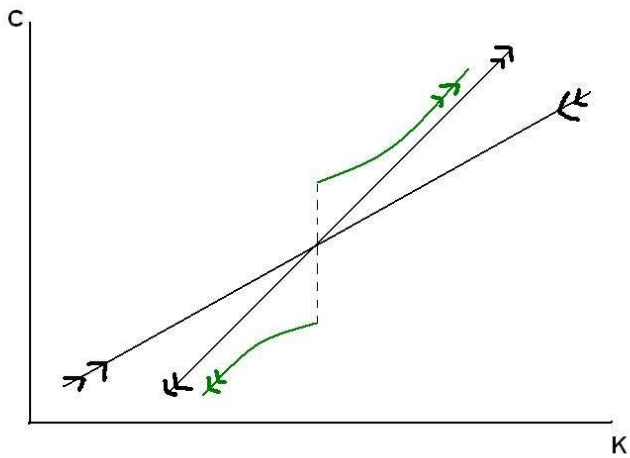
March 24, 2016

More on Challenge for Perception driven cycles

Illustrating the problem with phase diagrams

- ▶ Consider simple neo-classical setting: capital stock is the only state variable, and that technology and preferences are steady growth compatible.
- ▶ Dynamics can be represented in simple two dimensional phase diagram.





Comments

- ▶ Both stable arms are positively sloped $\rightsquigarrow$ same similar properties of the transitional dynamics (but not in growth rates)
- ▶ Very different behavior following expectations shocks.

More on two sector models (see Beaudry and Portier jet-2006)

- ▶ (i) Expectation driven fluctuations are difficult in the most commonly used macro models.
- ▶ (ii) If RA and CRS are maintained, then some internal cost complementarities are needed
- ▶ Not an exotic feature: implicitly implied by models which adopt inter-sectorial adjustment costs.

In a nutshell:

- We introduce the following setup:

$$E_0 \sum_{t=0}^{\infty} \beta^t U(C_t, 1 - L_t) \quad (1)$$

$$C_t = G(K_t, L_t, I_t; \psi_t) \quad (2)$$

$$K_{t+1} = I_t + (1 - \delta)K_t \quad (3)$$

In a nutshell:

$$C_t = G(K_t, L_t, I_t; \psi_t) \quad (2)$$

Remark : Equation (2) encompasses many of the macro models we use.

- ▶ One sector growth model $G(\cdot) = F(K_t, L_t) - I_t$, with $F(\cdot)$ concave
- ▶ Tobin's Q model $G(\cdot) = F(K_t, L_t) - I_t - \phi\left(\frac{I_t}{K_t}\right) K_t$, with $\phi(\cdot)$ a convex cost of adjustment function.

In a nutshell:

- ▶ Standard two-sector model: $G(\cdot)$ is the envelop function defined by

$$\begin{aligned} G(K, L, I) = \max_{K^c, K^I, L^c, L^I} \quad & F^c(K^c, L^c) \\ \text{s.t.} \quad & \\ & F^I(K^I, L^I) \geq I_t \\ & K^c + K^I \leq K_t \\ & L^c + L^I \leq L_t \end{aligned}$$

where $F^c(K^c, L^c)$ is the production function for consumption goods and $F^I(K^I, L^I)$ for investment goods.

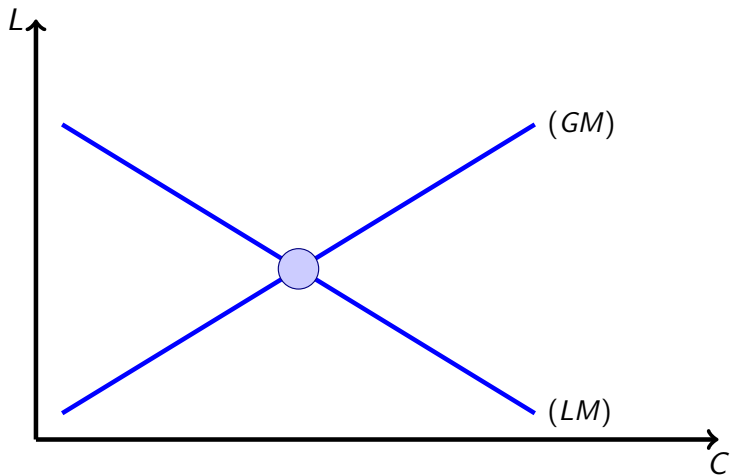
In a nutshell:

- ▶ We show that $G_{LI} > 0$ is a necessary condition for Expectation Driven Business Cycles (“Pigou Cycles”)
- ▶ Interpretation: cost complementarity.
- ▶ Labor is marginally more productive in the consumption good sector when the economy is investing a lot.
- ▶ Temporary equilibrium:

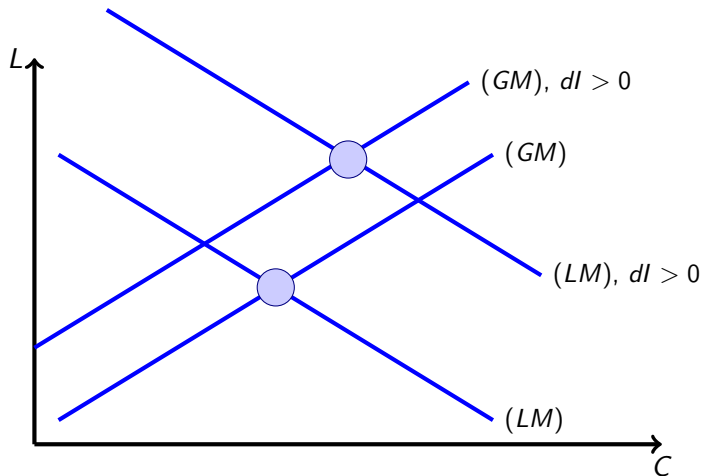
$$C = G(K, L, I) \quad (GM)$$

$$U_C(C, 1 - L)G_L(K, L, I) = -U_L(C, 1 - L) \quad (LM)$$

A Model with (large enough) Positive G_{LI}



A Model with (large enough) Positive G_{LI}



Departing from Representative agents

Basic assumption

- ▶ Two-agents/two-sector economy
- ▶ Maintaining CRS and separate production functions.
- ▶ Based on Beaudry-Portier, NBER macroannual 2013

- ▶ $U^i(C_i, 1 - L_i) + V^i(K_i; \Omega)$
- ▶ Concave CRS technologies in both sectors.
- ▶ $C = F^C(L_1^C, L_2^C)$ and $K = F^I(L_1^I, L_2^I)$.
- ▶ We will contrast:
 - ▶ integrated labor markets: $C = F^C(L_1^C + L_2^C)$ and $K = F^I(L_1^I + L_2^I)$.
 - ▶ (very) segmented labor markets: $C = F^C(L_1)$ and $K = F^I(L_2)$.

Perception Driven Fluctuations (News Shocks, etc)

- ▶ We study the competitive equilibrium of that economy.
- ▶ Under which conditions does a increase in the perceived marginal value of capital ($d\Omega_1 > 0$) does create a boom?
Does a decrease create a bust?

Perception Driven Fluctuations (News Shocks, etc)

Definitions

Definition 1 : **aggregate positive co-movement** : *aggregate consumption, aggregate investment and aggregate employment all strictly move in the same direction.*

Definition 2 : **individual positive co-movement**: *consumption, investment and employment levels for all individuals move in the same direction.*

Perception Driven Fluctuations (News Shocks, etc)

Walrasian equilibrium puts little restrictions on allocations

Proposition 1 : *Following a change in perceptions,*

- ▶ *Aggregate positive co-movements are possible,*
- ▶ *Individual co-movements are not possible.*

Perception Driven Fluctuations (News Shocks, etc)

The representative agent case

Corollary 1 : *With a representative agent, positive aggregate co-movement **are not possible**.*

Perception Driven Fluctuations (News Shocks, etc)

The importance of labor market segmentation

- ▶ What does matter for aggregate positive co-movements?
- ▶ Preference heterogeneity or labor market segmentation?

Perception Driven Fluctuations (News Shocks, etc)

The importance of labor market segmentation

Proposition 2 :

- ▶ *If labour markets are fully integrated, aggregate positive co-movement **are not possible**.*
- ▶ *If preferences are identical and labour markets not fully integrated, aggregate positive co-movement **are possible**.*

The Story

- ▶ Spain
- ▶ Two types of households
- ▶ Carpenters and Farmers
- ▶ Houses (capital good) and tomatoes (consumption good)
- ▶ In the short run, attachment to a sector

The Story

- ▶ The carpenter needs to eat, the farmer needs a shelter
- ▶ Static Gains from Trade between the two.
- ▶ Assume that the perceived value of houses decreases (bad news, pessimism)
- ▶ The relative price of houses in terms of tomatoes p will go down
- ▶ The carpenter will work less for two reasons
 1. he does not want as many houses as before
 2. he cannot trade as many houses as before
- ▶ L_I and I will go down.
- ▶ The farmer does not want to buy as many houses as before, and does not need to produce as many tomatoes for the trade: L_C and C .

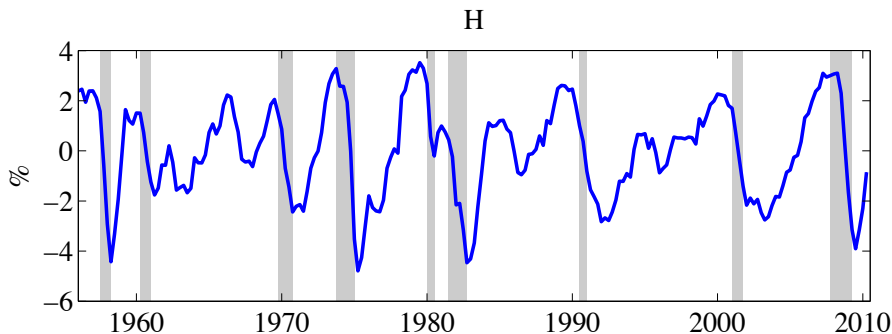
The Story

- ▶ C , I , L in both sectors, $Y = C + pI$ are all going down
- ▶ If reallocation of workers take some time, the recession is likely to be protracted.
- ▶ Changes in perceptions about the future are affecting the depth of Gains from Trade today
- ▶ Fluctuations are here related with variations in the amount of Gains from Trade.

Further motivation: Some Intriguing Facts over the last 3 cycles

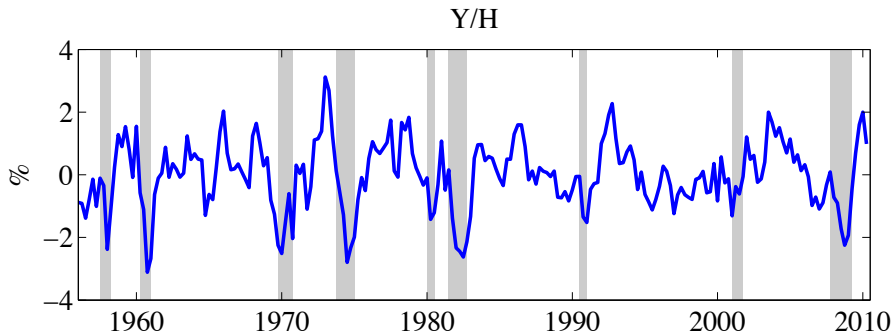
- ▶ Over the last 30, the US economy (can be extended to the Eurozone) has experienced an intriguing type of business cycle.
- ▶ Let's look at
 - ▶ HP filtered hours H
 - ▶ HP filtered average labor productivity Y/H
 - ▶ Raw core inflation

Some Intriguing Facts over the last 3 cycles



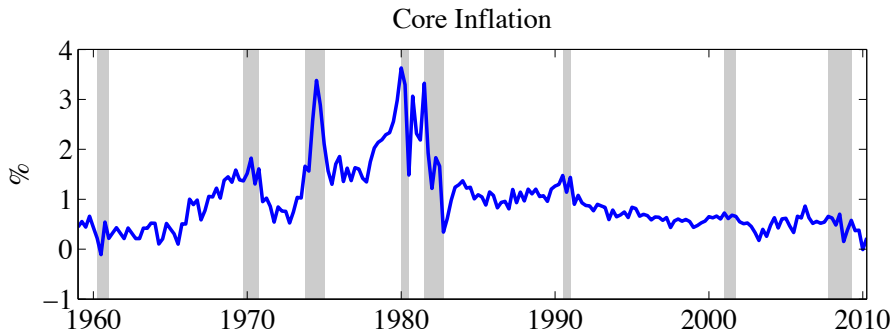
- ▶ The last (say) 3 business cycles do not look that different from the ones before.

Some Intriguing Facts over the last 3 cycles



- ▶ Productivity has been (at best) a-cyclical.
- ▶ This is pointing towards “demand” shocks as generating recent BCs.

Some Intriguing Facts over the last 3 cycles



- ▶ Core-inflation has been very stable.
- ▶ Almost no variance nor co-variance with output.

The Trouble with RBC Models

- ▶ This pattern is hard to explain in the RBC tradition :
- ▶ No procyclical TFP shocks.
- ▶ Demand shocks typically cause negative consumption - investment co-movements.

The Trouble with NK Models

- ▶ If output moves because the output gap moves (demand shocks), then we should expect inflation response.
- ▶ The core NK model has more than “troubles”
- ▶ Take the Jordi Galí’s textbook New Phillips curve

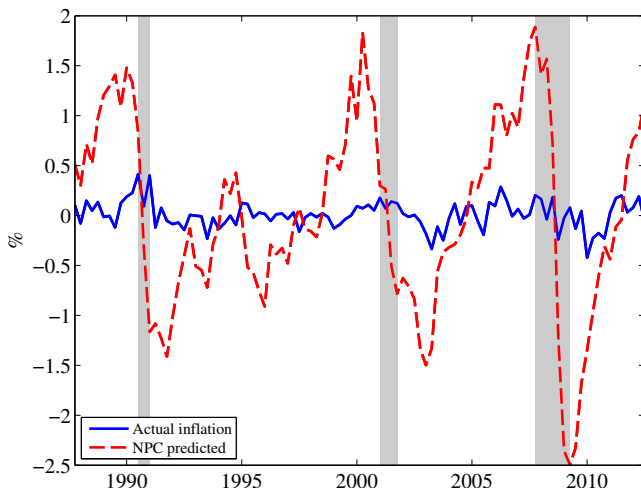
$$\pi_t = \beta E_t \pi_{t+1} + \kappa \tilde{y}_t + u_t$$

- ▶ Assume that the output gap is AR(1) with persistence ρ .

$$\pi_t = \frac{\kappa}{1 - \beta\rho} \tilde{y}_t + u_t$$

- ▶ Take Jordi’s textbook calibration (including a mean duration of prices of 3 quarters).
- ▶ Assume that the output gap is measured by the HP cycle of output (conservative assumption).
- ▶ Feed it into this last equation and deduct the implied inflation, killing cost-push shocks.

The Trouble with NK Models



- ▶ Then NPC implies that s.d. of inflation is **3** times larger than in the data

Introducing “Gains from Trade”

- ▶ Introduce our limited mobility / two sector framework within an otherwise standard New Keynesian model.
- ▶ Consider believes that future returns to capital will be high.
- ▶ Consumption producing agents want to by capital as it return is expected to be high.
- ▶ They will trade more with investment producing agents.
- ▶ Investment producing agents will also want more capital, but also more consumption because of an income effect.
- ▶ This would look like a demand shock.
- ▶ But it does not put any pressures on prices, as the natural or non-inflation rate of output has also changed.
- ▶ Same thing with downward revisions: recession but no deflation.

Introducing “Gains from Trade”

- ▶ We think it is a useful model to think of the last episodes of expansions and recessions.
- ▶ Productivity is not procyclical
- ▶ Inflation barely moves
- ▶ Suggest that booms and busts are mainly driven by expectations on investment return.

The basic model

- ▶ Let's start with the textbook New Keynesian model.
- ▶ $c_t = \left(\int_0^1 c_{jt}^{\frac{\varepsilon-1}{\varepsilon}} dj \right)^{\frac{\varepsilon}{\varepsilon-1}}$
- ▶ $\sum \beta^t (\ln(c_{ct}) + \Phi(1 - \ell_{Ct}))$.
- ▶ Monopoly $j : C_{jt} = A_t L_{jt}$
- ▶ Calvo price setting
- ▶ Monetary authorities follow a Taylor rule.

The basic model

The basic model

- ▶ This basic model collapses to a Phillips curve and a dynamic IS equation + Taylor rule

$$\left\{ \begin{array}{lcl} \tilde{y}_t & = & -(\hat{i}_t - E_t \hat{\pi}_{t+1} - \hat{\rho}_t^n) + E_t \tilde{y}_{t+1} \\ \hat{\pi}_t & = & \beta E_t \hat{\pi}_{t+1} + \lambda (\hat{y}_t - \hat{y}_t^n) \\ + & & \text{Taylor rule} \end{array} \right.$$

- ▶ $\hat{y}_t^n = \hat{A}_t$

A GFT approach

The basic model

$$\begin{cases} \tilde{y}_t &= -(\hat{v}_t - E_t \hat{\pi}_{t+1} - \hat{\rho}_t^n) + E_t \tilde{y}_{t+1} \\ \hat{\pi}_t &= \beta E_t \hat{\pi}_{t+1} + \lambda (\hat{y}_t - \hat{y}_t^n) \\ + &\text{Taylor rule} \end{cases}$$

- ▶ What happens in this environment if agents expect $\hat{A}_{t+1}$ to be high?
- ▶ It is a demand shock to current consumption.
- ▶ Absent of an active Taylor Rule, inflation.
- ▶ The increased expectation of A_{t+1} does not directly enter into the Phillips curve.
- ▶ Therefore, if it creates an output boom, it will create inflation (no change in the natural output)

A GFT approach

The basic model

$$\left\{ \begin{array}{lcl} \tilde{y}_t & = & -(\hat{v}_t - E_t \hat{\pi}_{t+1} - \hat{\rho}_t^n) + E_t \tilde{y}_{t+1} \\ \hat{\pi}_t & = & \beta E_t \hat{\pi}_{t+1} + \lambda (\hat{y}_t - \hat{y}_t^n) \\ + & & \text{Taylor rule} \end{array} \right.$$

- ▶ What happens in this environment if agents observe high $\hat{A}_t$?
- ▶ It is a supply shock .
- ▶ It shows up directly in the the Phillips curve ($\hat{y}_t^n = \hat{A}_t$).
- ▶ Therefore, if it creates an output boom, it will not create inflation (changes in the natural output)

A GFT approach

The basic model

- ▶ Textbook prescription: to keep stable inflation, monetary authorities need to strongly counteract demand shock but need to accommodate supply shocks.
- ▶ If the economy goes into recession due to a fall in demand – as opposed to a reduction in supply capacity – this should put substantial downward pressure on prices.
- ▶ Let's contrast those results with the one of a gains-from-trade model.

A GFT approach

The gains-from-trade model

- ▶ Add a mass 1 of agents (type 2)
- ▶ Type 2 individuals produce some capital good (full depreciation): $K_{t+1} = L_{2t}$.
- ▶ Preferences: $\ln \left(C_{2t} + \frac{\psi_2}{2}(1 - L_{2t})^2 \right)$.
- ▶ Capital market is flex-price and competitive.
- ▶ Production of consumption good: $C_{jt} = \Theta_t K_{jt} + L_{1jt}$
- ▶ If then mass of type 2 individuals is 0, then there is no capital supply and the model is the previous one.

A GFT approach

The gains-from-trade model

- ▶ The log linear approximation of the model is

$$\left\{ \begin{array}{lcl} \tilde{y}_t & = & -\zeta (\hat{i}_t - E_t \hat{\pi}_{t+1} - \hat{\rho}_t^n) + E_t \tilde{y}_{t+1} \\ \hat{\pi}_t & = & \beta E_t \hat{\pi}_{t+1} + \lambda \zeta^{-1} (\hat{y}_t - \hat{y}_t^n) \\ + & & \text{Taylor rule} \end{array} \right.$$

- ▶ Natural or non-inflationary level of output

$$\hat{y}_t^n = \phi_2 A_t + \phi_1 E_t [\hat{\Omega}_{t+1}] \text{ where } \hat{\Omega}_{t+1} = \hat{\Theta}_{t+1} - \hat{A}_{t+1}$$

A GFT approach

A new new Phillips curve

$$\hat{\pi}_t = \beta E_t \hat{\pi}_{t+1} + \lambda \zeta^{-1} \left(\hat{y}_t - \phi_2 A_t - \phi_1 E_t \left[\hat{\Omega}_{t+1} \right] \right)$$

- ▶ Consider believes that Ω_{t+1} will be high.
- ▶ Agents of type 1 to want to by capital as it return is expected to be high.
- ▶ They will trade more with agents of type 2.
- ▶ Type 2 agents will also want more capital, but also more consumption because of an income effect.
- ▶ This would look like a demand shock.
- ▶ But it does not put any pressures on prices, as the natural or non-inflation rate of output has also changed.
- ▶ Same thing with downward revisions: recession but no deflation.

Evidence of Labour Market Segmentation and Imperfect Insurance

Main idea

- ▶ Our theory relies on two main features of the world
 - ▶ (1) Sluggish sectoral reallocation of labor
 - ▶ If not, perceived changes in the value of capital would cause reallocations, not changes in the depth of gains to trade.
 - ▶ (2) Imperfect risk sharing
 - ▶ If not, consumption is immune to changes in expectations
- ▶ Is it possible to test for this dependence using PSID
- ▶ Idea: under full risk sharing and labor mobility, the beginning of period t sector of employment of an individual should not influence, *ceteris paribus*, the growth of her wage income or consumption between, say, t and $t + n$.

Evidence of Labour Market Segmentation and Imperfect Insurance

Data

- ▶ Data from the PSID over the period 1968-2007 (yearly until 1997, bi-annual since).
- ▶ We observe for the family head and for every family-year
 - ▶ sector of activity at the beginning of the period
 - ▶ labour income
 - ▶ food consumption
 - ▶ age, education
- ▶ We also observe employment and wage bill at the sectoral level (using NIPA)

Evidence of Labour Market Segmentation and Imperfect Insurance

Specification

- ▶ We estimate:

$$\Delta X_{it} = \text{time dummies} + \alpha_1 \Delta Emp_{j(i)t} \\ + \text{age dummies} + \text{sector dummies} + \epsilon_{it}$$

- ▶ X is first labor income, then consumption.
- ▶ $j(i)$ is the sector in which i was employed at the beginning of the period.
- ▶ Δ is taken over 2 years, or 1 year after 1997
- ▶ Other controls: highest level of educational attainment, and interactions between age-time, education-age, education-time.
- ▶ If labour markets were completely integrated, and given we are controlling for common time effects then individual level outcomes should not be systematically related to aggregate outcomes for any particular sectors.

Evidence of Labour Market Segmentation and Imperfect Insurance

Table 1: Effect of Sectoral Growth on Individual Income

	2-year (1)	2-year (2)	2-year (3)	2-year (4)	2-year (5)	1-year (6)	1-year (7)	1-year (8)
Δ Emp	.542 (.209)		.468 (.244)					
Δ W-bill		.525 (.175)						
Δ Emp-10				.450 (.143)	.563 (.131)			
Δ Emp						.535 (.170)	.579 (.193)	
Δ Emp-10								.471 (.059)
Obs.	49338	49338	45469	45430	23173	68863	63677	61224
R^2	.028	.028	.028	.027	.026	.017	.018	.018

8.6. Evidence of Labour Market Segmentation and Imperfect Insurance

Table 1: Effect of Sectoral Growth on Individual Income

- ▶ Compare two individuals that were initially attached to two different sectors,
- ▶ assume one of the two sector grew by an extra 1% over two years,
- ▶ the individual initially attached to that sector saw his labour income grow by an additional .5% compared to the other individual.

Evidence of Labour Market Segmentation and Imperfect Insurance

Table 2: Effect of Sectoral Growth on Household Consumption (Food)

	2-year (1)	2-year (2)	2-year (3)	2-year (4)	2-year (5)	1-year (6)	1-year (7)	1-year (8)
Δ Emp	.268 (.092)		.267 (.104)					
Δ W-bill		.236 (.078)						
Δ Emp-10				.143 (.052)	.112 (.053)			
Δ Emp						.200 (.118)	.274 (.129)	
Δ Emp-10								.208 (.077)
Obs.	67758	67758	63686	52270	26898	89008	83942	65503
R^2	.014	.014	.013	.016	.015	.005	.005	.006

9. Fiscal Policy

Gains from trade as a foundation for multipliers

- ▶ Typical story for multipliers in keynesian-cross-like-models:
 - ▶ Initial increase in demand for one good,
 - ▶ This increase in demand is met by an increase in production of that good, and therefore by an increase in income of those producing that good
 - ▶ Those agents then in turn increase demand for some goods, etc...
- ▶ This is not (at all) the mechanism we find in sticky prices models of the business cycle.
- ▶ Let us show that multipliers can be understood as the consequence of changes in the scope for gains to trade between agents.

9. Fiscal Policy

Framework

- ▶ We again use the simple model ($U^i(C_i, 1 - L_i)$ and $V^i(K_i; \Omega_1, \bar{K})$, two sectors)
- ▶ We abstract for Ricardian issues, and look at balanced-budget policies of the type

$$G^C + G^K = T_1 + T_2$$

- ▶ Under what conditions (if any) can a temporary increase in government spending create a multiplier effect?
- ▶ It is helpful to begin by focusing on government expenditures that are directed only at one sector.

$$G^K = T_1 + T_2$$

9. Fiscal Policy

Two questions

- ▶ Given a policy (G^K, T_1) (T_2 is then obtained from the govt BC), two questions can be asked:

Question 1 : amplification effect: *Can an increase in government purchases of capital good cause private agents to buy more capital goods?*

Question 2 : spillover effect: *Can an increase in government purchases of capital good cause private agent to consume more consumption goods?*

9. Fiscal Policy

Two propositions

Proposition 5 :

- ▶ *If same preferences and perfectly substitutable labour,*
- ▶ *no increase in C (spillover) nor I (amplification) are possible.*

9. Fiscal Policy

Two propositions

Proposition 6 :

- ▶ *If the conditions of Proposition 3 are met,*
- ▶ *C and total I will increase.*
- ▶ *However, no amplification (private I decreases)*

9. Fiscal Policy

Two propositions

- ▶ An increase in government purchases of capital shifts out the aggregate demand for capital, increasing the aggregate quantity purchased and increasing the price of capital.
- ▶ Individual demand curves for capital tend to shift in because of the tax increases.
- ▶ No amplification effect as the equilibrium price of capital increase and agents have downward sloping demand curves.
- ▶ Agents in the capital sector are getting higher income which leads them to want to consume more. This is where the gains from trade arise.
- ▶ The producers of consumption goods are willing to trade with the producers of capital goods as the added income for them can be used to pay taxes.

9. Fiscal Policy

Two propositions

- ▶ Could it be the case that the income effect on the producers of capital is great enough to lead such producers to actually substantially increasing their purchase of capital and create amplification?
- ▶ It can happen, but not under the conditions of Proposition 3 (aggregate capital demand is downward sloping).

9. Fiscal Policy

An Example

- ▶ We keep the same two periods environment:
In $C^i + \phi(1 - I^i) + \ln K^i$, $C = L_1$, $K = L_2$, P the relative price of capital.
- ▶ Fiscal policy: spend α on K (amount spent $\frac{\alpha}{P}$), $1 - \alpha$ on C
- ▶ Fiscal policy: tax β on type 2 agent, $1 - \beta$ on type 1.
- ▶ In equilibrium:
 - ▶ $GDP = \frac{4}{\phi}$ without spending
 - ▶ $GDP = \frac{4}{\phi} + 1 + 2(\alpha - \beta)$ when spending 1.
- ▶ Multiplier = $1 + 2(\alpha - \beta)$

Conclusion

- ▶ Lot of things to do : asset pricing, endogenize the acquisition and filtering of information
- ▶ À suivre...